

Context Vector Analysis in LLMs

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Relevant Code: [\[Colab\]](#)

1. Dataset & Model

Dataset: The *Pride and Prejudice* Book dataset was utilized, from which 1800 sentences by using sentence tokenizer.

Model: A pretrained “bert-base-uncased” model was used.

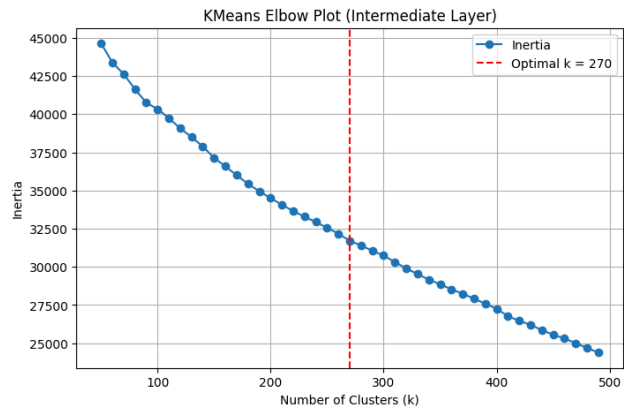
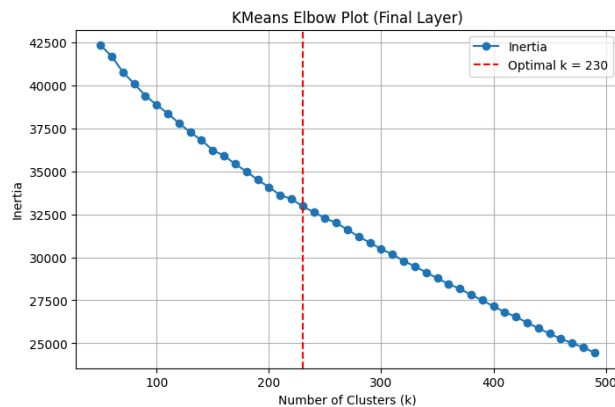
2. Grouping Method

To group sentences, four different clustering approaches were explored: (1) **greedy algorithm**, (2) **graph-based grouping**, (3) **DBSCAN**, and (4) **K-means clustering**. All methods aim to avoid assigning duplicate sentences into the same group. For cosine similarity based on approaches, like greedy, graph-based, and DBSCAN ($\epsilon = 1 - \cos_thres$), a threshold of 0.9 was selected, given that the average cosine similarity between first sentences is relatively high.

As observed, both Graph-based and DBSCAN methods tended to produce large clusters with many singleton groups, indicating potential noise and imbalance. Hence, the **Greedy algorithm** among first third methods was selected as the primary method for sentence grouping.

```
--Greedy Grouping--
Final layers' Group Counts of first 5th group: [11, 3, 3, 14, 78]
Intermediate layers' Group Counts of first 5th group: [228, 12, 7, 27, 2]
--Graph Grouping--
Final layers' Group Counts of first 5th group: [1641, 1, 1, 1, 1]
Intermediate layers' Group Counts of first 5th group: [1734, 2, 1, 1, 6]
--dbscan Grouping--
Final layers' Group Counts of first 5th group: [1641, 136, 2, 2, 3]
Intermediate layers' Group Counts of first 5th group: [1734, 2, 41, 6, 3]
```

Additionally, to determine the effectiveness of K-means clustering, elbow analysis was performed to select the optimal number of clusters ($n_clusters$). The range of k was set from 50 to 500 with a step size of 10. As shown in the elbow plots below, the optimal values were approximately: 230 for final layer embeddings and 270 for intermediate layer embeddings. These optimal values were chosen based on the elbow point detected by the KneLocator algorithm.



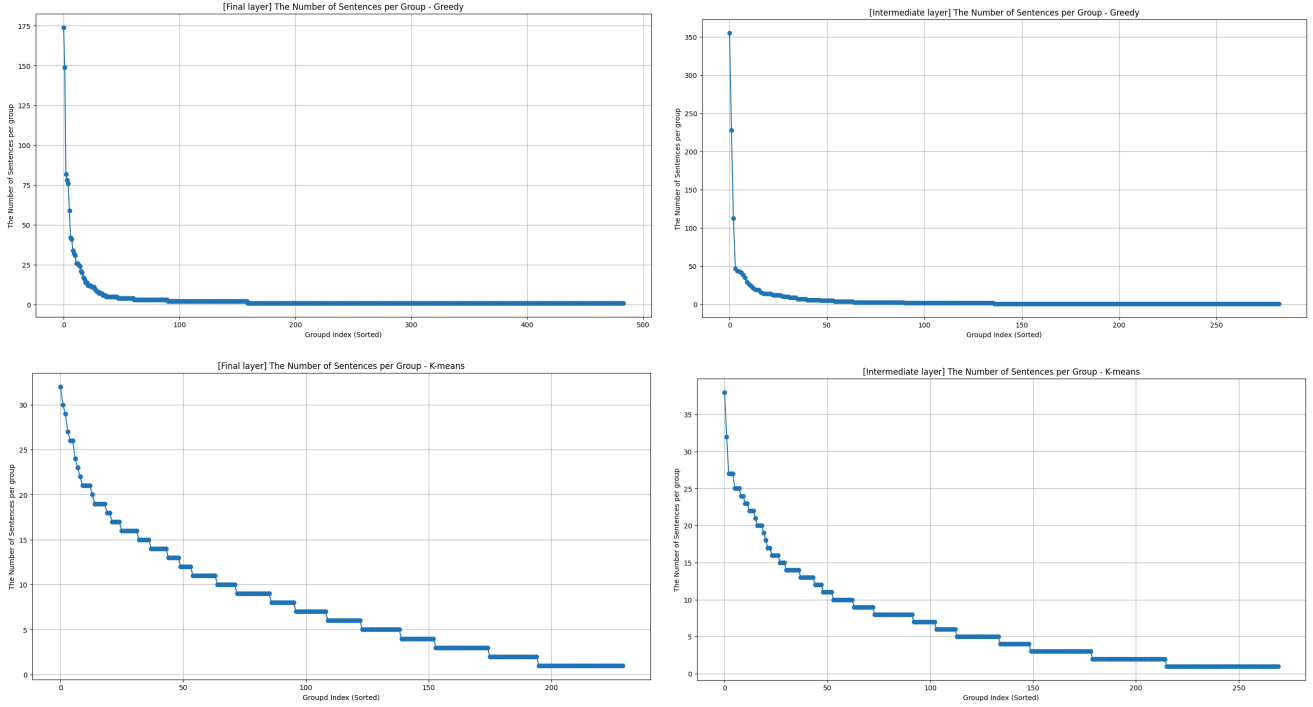
```
--K-means Grouping--
Final layers' Group Counts of first 5th group: [32, 30, 29, 27, 26]
Intermediate layers' Group Counts of first 5th group: [38, 32, 27, 27, 27]
```

To ensure that no sentence duplication occurred during grouping, a sanity check was conducted. The total number of sentences assigned across all clusters matched the original sentence count:

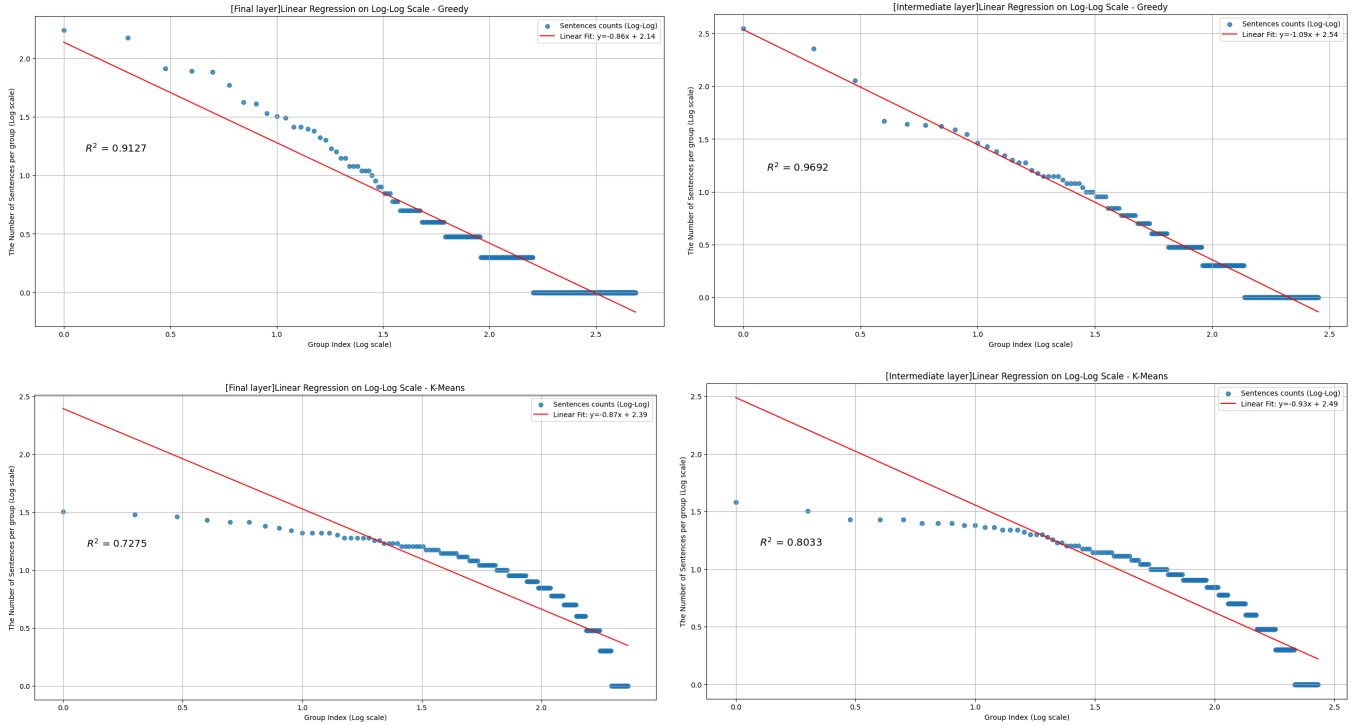
```
sanity check using the number of sentences in each group
The total number of sentences in the Final layers: 1800
The total number of sentences in the Intermediate layers: 1800
Sanity check result: True
```

3. Result and Analysis

To examine internal clustering behavior within BERT, both final and intermediate layer embeddings were analyzed. For the intermediate representation, the 8th layer of BERT (i.e., 2nd/3rd from the beginning) was selected to serve as the intermediate context vector. Based on these embeddings, I evaluated how sentences grouped together using two clustering strategies: **Greedy** and **K-means clustering**. Following Zipf's Law, the number of sentences in each group was sorted in decreasing order and visualized.



Greedy Clustering yielded the following slopes: **-0.86** for Final layer and **-1.09** for intermediate layer. Also, K-means clustering yielded: **-0.87** for Final layer and **-0.93** for intermediate layer.



Both methods produced similar slopes in the final layer (-0.86 vs. -0.87), implying consistent grouping characteristics regardless of clustering strategy at this stage. Additionally, the intermediate layer exhibits stronger Zipf-like behavior than the final layer, possibly due to more generalized or abstracted semantic representations. These findings suggest that intermediate layers may reveal more hierarchical or structured clustering behavior, while final layers may be more specialized toward task-specific distributions.